

Instructions



How is it going ? Do you have problems with practical tasks ? Any concerns with learning materials ?

no unit tests in task

no tests executed

heeeelp good

autocode error

tests are not configured

error starting autocode

no questions

problems i had not

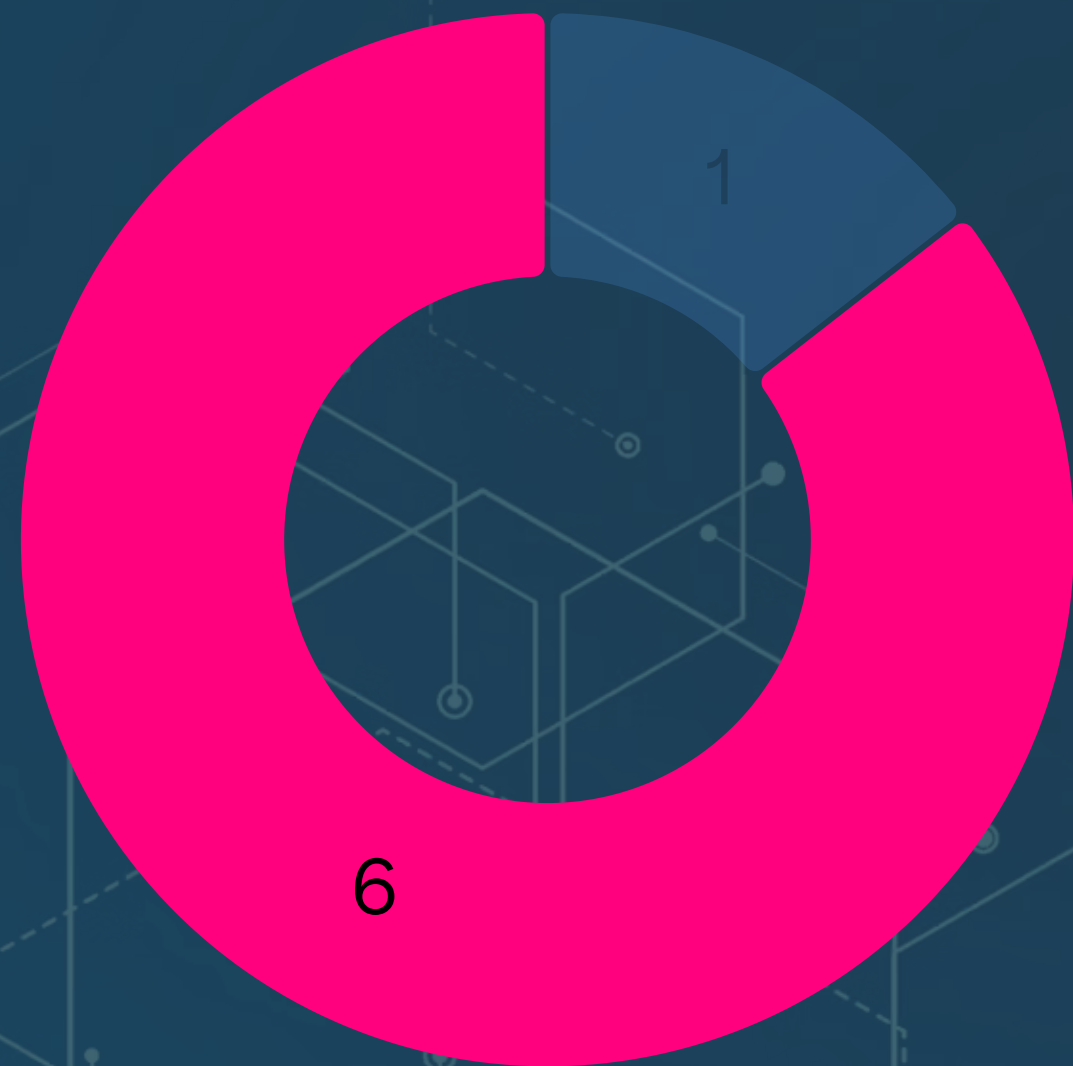
Today's agenda

- Total Recall
- Hibernate Cache
- Quiz
- Questions

Total Recall



What is TRUE about @Transactional annotation in JPA?



1

All operations within a method will be committed to database



0

It auto creates new instances of entities in the persistence context

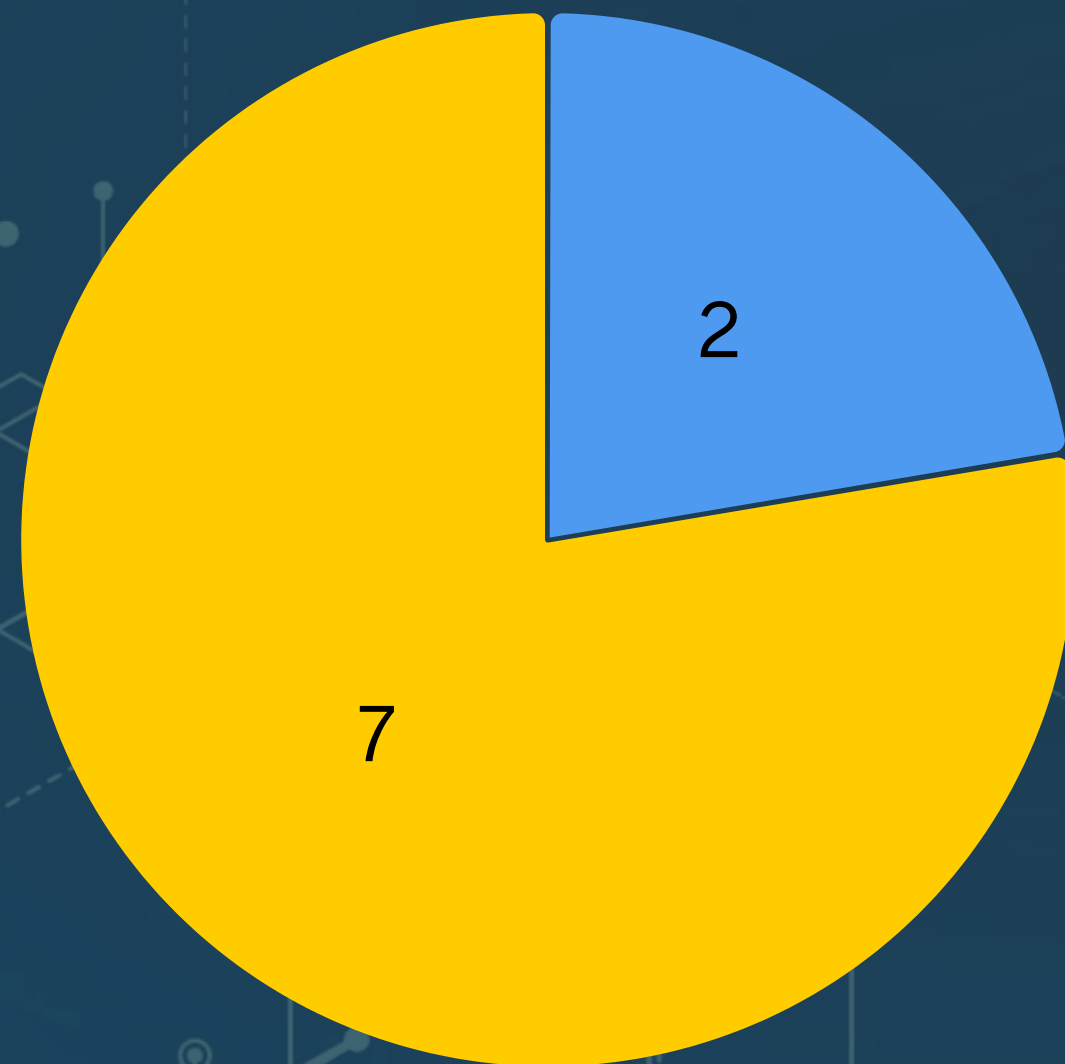


6

Method will run in scope of a transaction and finish with a commit or a rollback



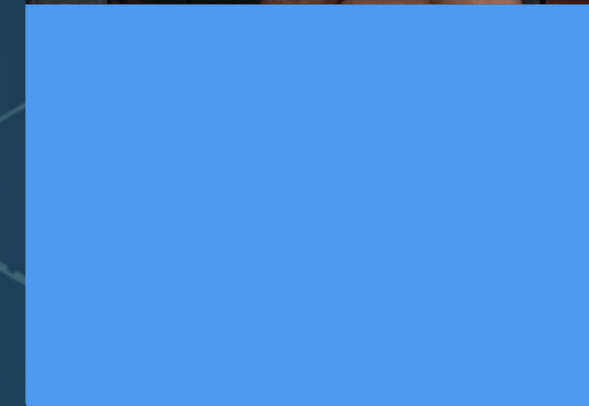
Where can we place @Transactional annotation ?



2	Above the class	✓
7	Above the method	✓
0	Above the code block {}	✗

Can we state that EntityManager is the same as Session, while leveraging Hibernate?

4 ✓



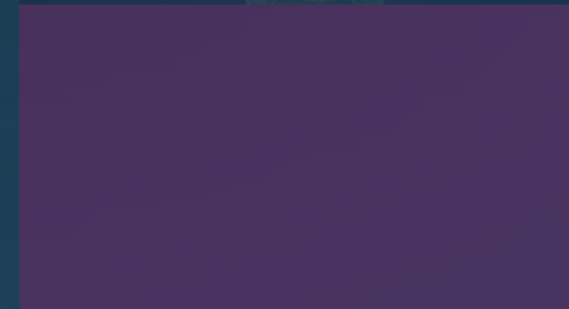
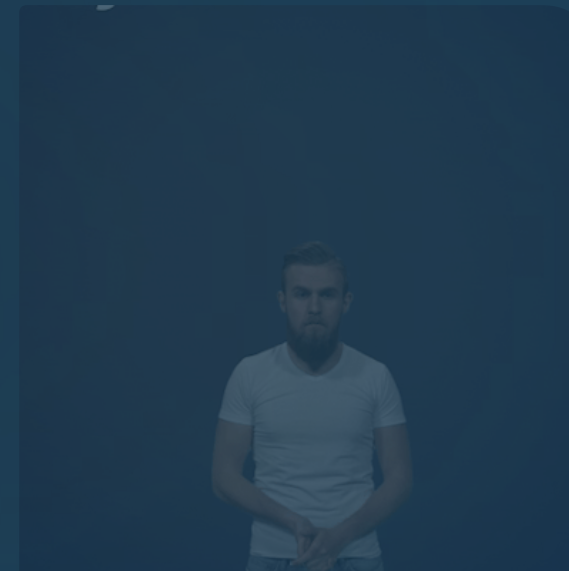
Oh yes we can

4 ✗



Not the same

3 ✗



Who knows

How can we mark a field which should be ignored in JPA?

10 



0 

Just leave the field as is

0 

Put an @Transient above

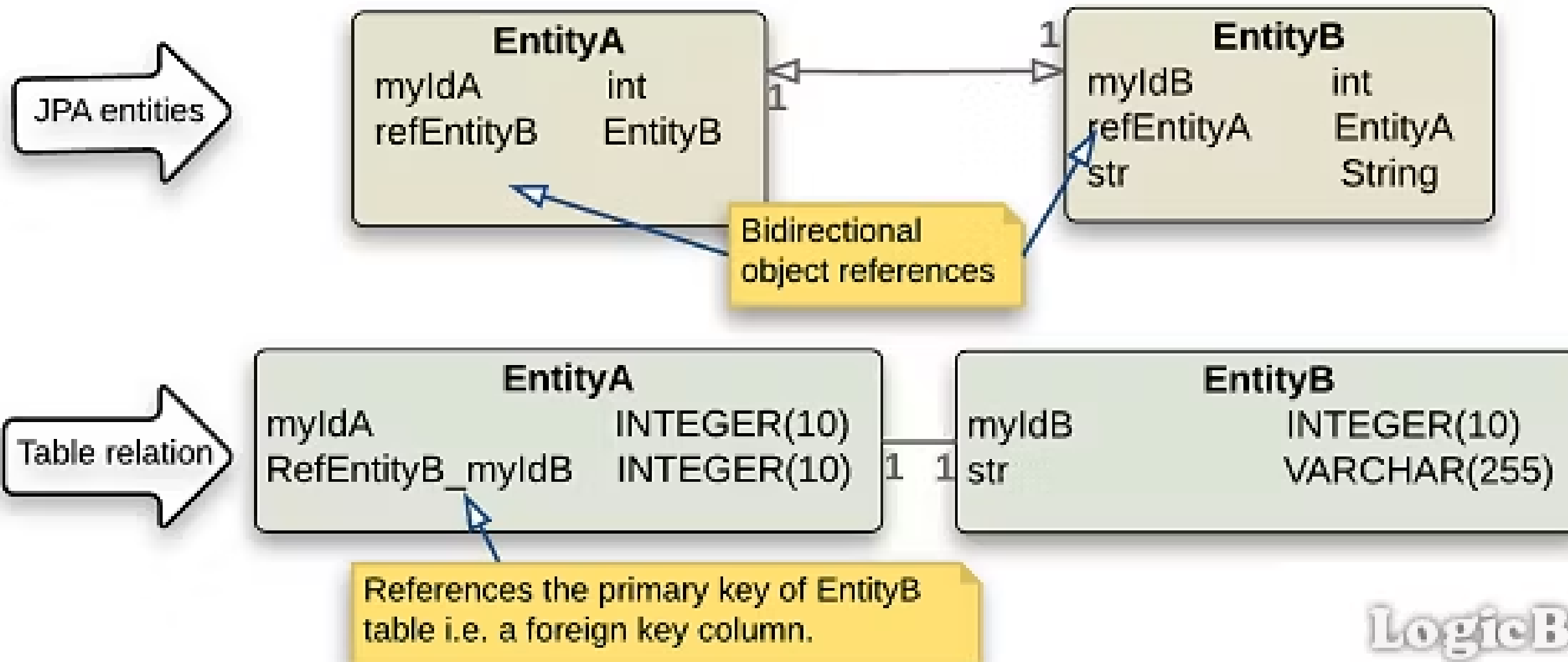
Make the field static

What are '*bidirectional relationship*' and '*unidirectional relationship*' in JPA ?

unidirectional - a and b
both entities reference

bidirectional - a owns b
only one entity

One-to-One Bidirectional Relationship



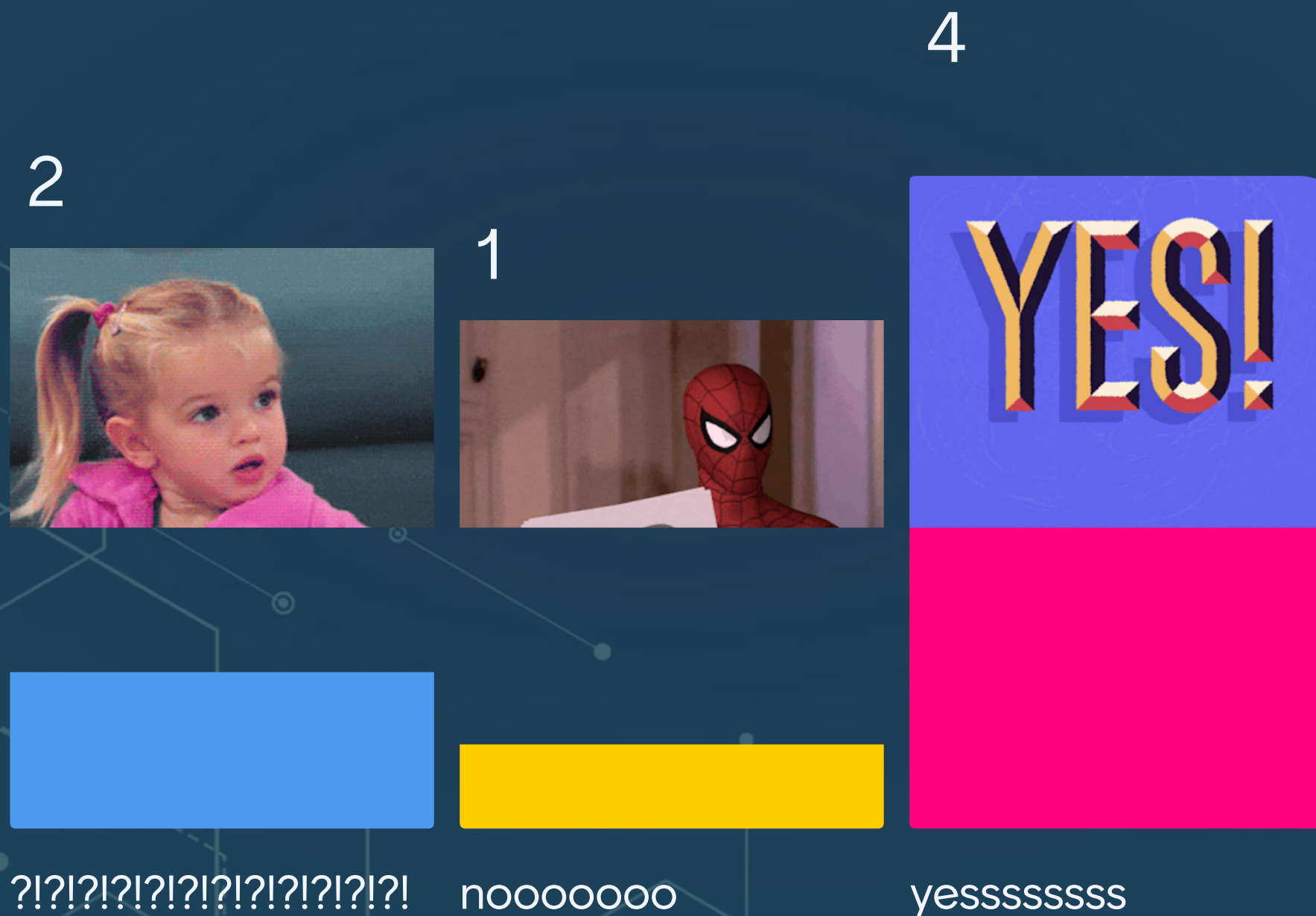
<https://www.logicbig.com/tutorials/java-ee-tutorial/jpa/one-to-one-bidirectional.html>

Hibernate (ws-08)

Normalization | Isolation | Transaction | Query



Have you started the theory of 'Hibernate'?



ACID: <https://en.wikipedia.org/wiki/ACID>

Second level cache

How it works:

When hibernate session tries to load an entity, it will first check it in first-level cache, if data is not found then it will look into the second-level cache and return the response (if available), but before returning the response it will store that object/data in first-level cache also - so next time no need to check in session-level.

When data is not found in the second level then it will address database to fetch data. Before returning a response to the user it will store that object/data into both levels of cache so next time it will be available at cache stages.

Explore module ws-08

<https://github.com/AndreiRohau/itpu-web-development-with-java-2026>

More about cache

<https://dzone.com/articles/pitfalls-hibernate-second-0>

Explain N+1 problem in Hibernate

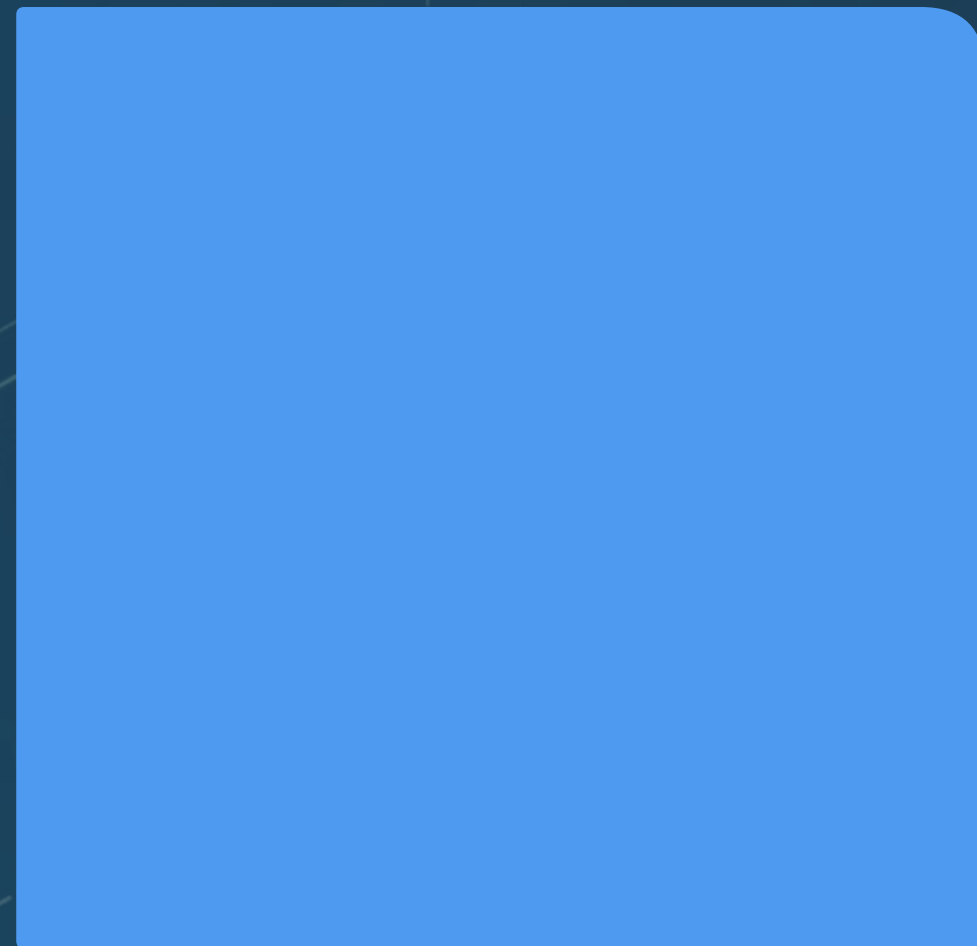
<https://github.com/mjc-school/MJC-School/blob/main/stage%20%233/module%20%233.%20ORM/lesson%20%237.%20N%2B1%20problem%20in%20Hibernate/content.md>

10 mins beak

Quiz

Can we wrap and unwrap EntityManagerFactory and EntityManager to get access to SessionFactory and Session ?

19 ✓



yes

1 ✗

no

0 ✗

no idea

How many caches are there in Hibernate?

14 



2

5 



3

0 

4

Whom does second level cache belong to ?

18 ✓



0 ✗

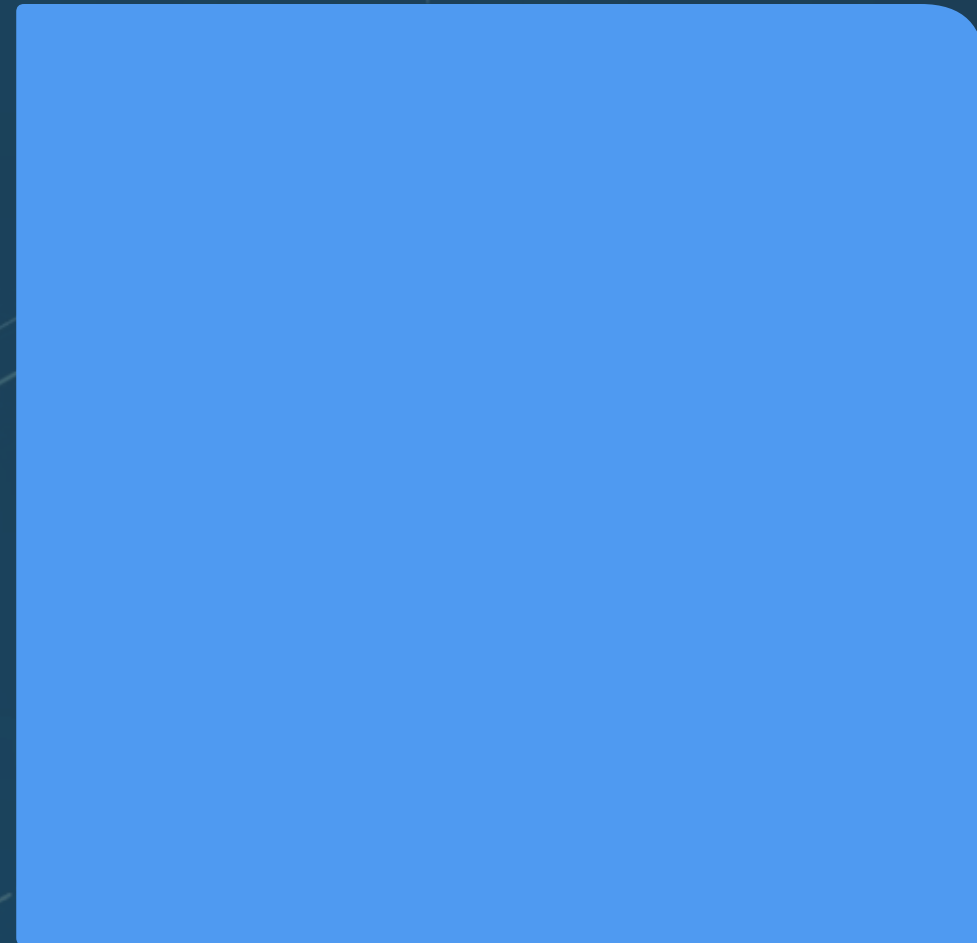
context of entityManager

1 ✗

context of second level cache provider

Hibernate does NOT provide *default* implementation for the second-level cache

21 ✓



true

0 ✗

false

0 ✗

no idea

How to *enable* 2nd level cache ?

1 ✕

Just leverage proper annotations

0 ✕

It's working all the time

21 ✓

Enable it in config file

How to enable 3rd level cache ?

2 ✕

Using annotations

1 ✕

In config file

19 ✓

There is no such

query cache

3rd level cache does not exist, but there is a *query cache*, which can be enabled by setting *hibernate.cache.use_query_cache* property to *true*

Where can we put this line: `@Cache(usage=CacheConcurrencyStrategy.READ_WRITE)`

1 ✗

above constructor

0 ✗

above method

21 ✓

above class and field

What is stored in Query Level Cache ?

0 ✕

database snapshot

0 ✕

table snapshot

24 ✓

query data and results

N+1 problem in Hibernate is when...

22 ✓



1 ✗

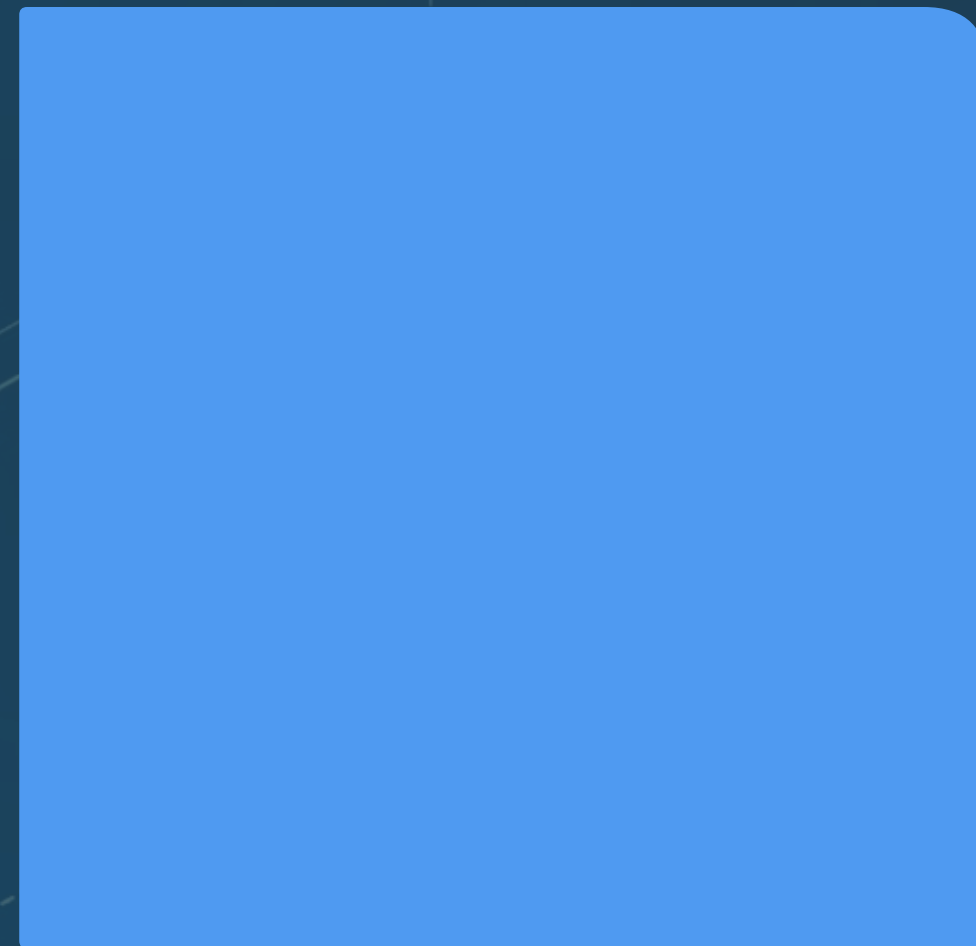
during persistence a duplication is saved

1 ✗

you have to add second level cache dependency library

Approach to solve n+1 problem is:

19 ✓



use eager fetching

2 ✗



use lazy load

3 ✗



use bidirectional dependency

Quiz leaderboard



Discuss Hibernate practical task

Next time: HTTP

Questions?

